

FITBUDDY

AI Fitness Plan Generator using Gemini Models

GROUP PROJECT DOCUMENTATION

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Course	BCA
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Project Title	FitBuddy – AI Fitness Plan Generator using Gemini Models
Technologies	Python, FastAPI, Jinja2, HTML, CSS, Gemini AI, SQLite / SQLAlchemy

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1. Project Overview

FitBuddy is a web-based AI fitness planning application designed to generate personalized workout guidance from user-provided information. The system accepts details such as name, age, weight, fitness goal and workout intensity, then uses a Gemini model to produce a structured plan. The project also supports feedback-based plan refinement and an administrative view of stored user information.

The application uses FastAPI for backend routing and request handling, Jinja2 for dynamic HTML pages, and SQLite/SQLAlchemy for storing application data. The reference project describes pages such as the home page, personalized result page, feedback page and view-all-users page.

Main Objectives:

- Generate a personalized 7-day workout plan.
- Use AI to tailor the output to the user's fitness goal and selected intensity.
- Provide simple nutrition or recovery guidance.
- Allow users to submit feedback for plan improvement.
- Store and display relevant user and plan information.

2. Brainstorming & Ideation

The idea for FitBuddy was developed around the need for a simple fitness assistant that can generate personalized guidance rather than giving the same plan to every user. The team identified AI-based personalization, an easy web interface, feedback, and stored user information as useful features.

3. Requirement Analysis

The functional requirements include collecting user details, generating a workout plan, displaying the generated result, accepting feedback, updating the plan where supported, and providing an administrative/user-list view. The non-functional requirements include usability, reasonable response time, maintainable code, and protection of sensitive configuration such as the Gemini API key.

4. Project Design

The system follows a simple web architecture. The browser sends form data to the FastAPI backend. The backend validates and processes the input, communicates with the Gemini service, stores relevant data through the database layer, and renders a Jinja2 template containing the result.

5. Project Planning

The project can be organized into milestones: selecting the AI model and architecture, developing backend functionality and routes, creating the frontend templates, connecting the database, testing the complete flow, and preparing the application for local demonstration. The reference project similarly organizes work into model/architecture, core development, route development, frontend development and deployment milestones.

6. Project Development

During development, the FastAPI application handles routes and user input. Jinja2 templates provide the web pages. The reference structure identifies `index.html` for collecting input, `result.html` for displaying the generated AI plan, and `all_users.html` for the administrative view. Environment variables are used for the Gemini API key rather than placing the key directly in source code.

7. Project Testing

Testing is performed by checking the main user flow from entering details to receiving a generated result. Form validation, successful AI response, result-page rendering, feedback submission, database storage, and the all-users view should be checked before demonstration.

8. Project Documentation

The documentation explains the purpose of FitBuddy, system requirements, architecture, development process, testing approach, user interface pages, source-code highlights and final result. Screenshots from the group's actual running application should be inserted in the screenshot section before submission.

9. Project Demonstration

For the demonstration, the team can show the home page, enter sample user details, generate a personalized plan, display the result, demonstrate feedback if implemented, and show the stored-user/admin page if available. The reference project describes the home page and personalized workout result as important parts of the demonstration.

10. Source Code - main.py

```
import os
from fastapi import FastAPI, Request, Form
from fastapi.templating import Jinja2Templates
from fastapi.responses import HTMLResponse

app = FastAPI(title="FitBuddy AI Fitness Plan Generator")
templates = Jinja2Templates(directory="templates")

@app.get("/", response_class=HTMLResponse)
async def home(request: Request):
    return templates.TemplateResponse(
        "index.html", {"request": request}
    )

@app.post("/generate", response_class=HTMLResponse)
async def generate(
    request: Request,
    name: str = Form(...),
    age: int = Form(...),
    weight: float = Form(...),
    goal: str = Form(...),
    intensity: str = Form(...)
):
    # Connect this section to the Gemini service in your project.
    workout_plan = "AI-generated personalized workout plan"
    return templates.TemplateResponse(
        "result.html",
        {
            "request": request,
            "name": name,
            "age": age,
            "weight": weight,
            "goal": goal,
            "intensity": intensity,
            "workout_plan": workout_plan
        }
    )
```

Note: This is a documentation-friendly code extract. Keep your group's actual working main.py code in the final project folder.

11. Source Code - result.html

```
<!DOCTYPE html>
<html>
<head>
    <title>FitBuddy Result</title>
</head>
<body>
    <h1>Your Personalized Fitness Plan</h1>
    <p><b>Name:</b> {{ name }}</p>
    <p><b>Age:</b> {{ age }}</p>
    <p><b>Weight:</b> {{ weight }} kg</p>
```

```
<p><b>Goal:</b> {{ goal }}</p>
<p><b>Intensity:</b> {{ intensity }}</p>

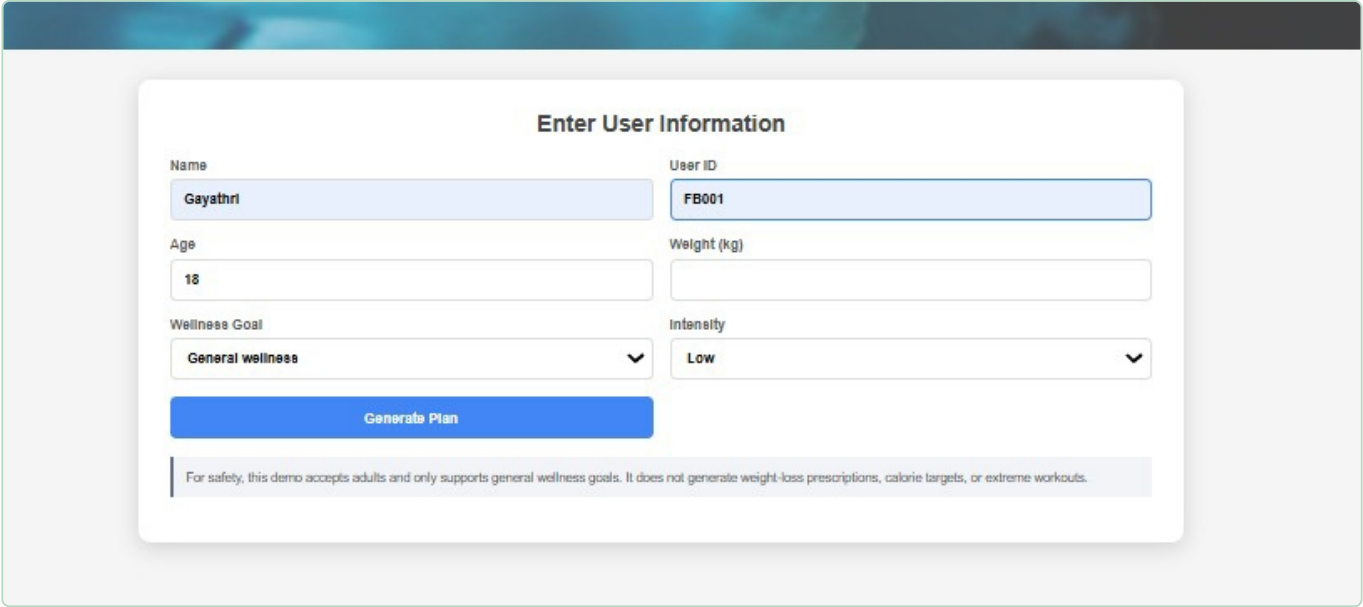
<h2>Workout Plan</h2>
<div>
    {{ workout_plan }}
</div>
</body>
</html>
```

Note: Replace this extract with the exact result.html code used by your group if your file has additional styling or fields.

12. Project Screenshots

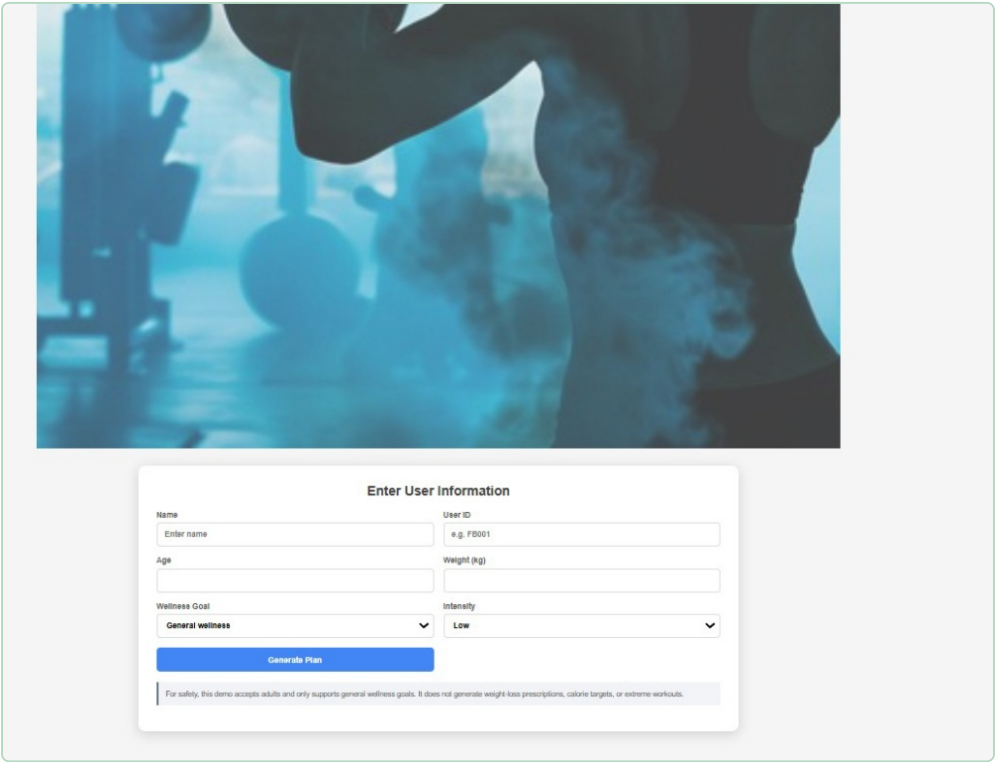
Screenshots from the running FitBuddy application demonstrating the full user interaction flow:

Screenshot 1 & 2 - FitBuddy Home Page & User Details Form



The screenshot displays the 'Enter User Information' form, which is centered on a light gray background. The form has a white header with the title 'Enter User Information'. It contains several input fields: 'Name' (with the value 'Gayathri'), 'User ID' (with the value 'FB001'), 'Age' (with the value '18'), 'Weight (kg)' (empty), 'Wellness Goal' (a dropdown menu with 'General wellness' selected), and 'Intensity' (a dropdown menu with 'Low' selected). A blue 'Generate Plan' button is located below the form fields. At the bottom of the form, there is a small disclaimer: 'For safety, this demo accepts adults and only supports general wellness goals. It does not generate weight-loss prescriptions, calorie targets, or extreme workouts.'

Screenshot 3 - Generated Workout Result



The screenshot shows the 'Generated Workout Result' page. It features a large, blurred image of a person's silhouette in a gym setting, with a blue tint. Below the image is the 'Enter User Information' form, which is identical to the one in Screenshot 1. The form fields are: 'Name' (with the value 'Enter name'), 'User ID' (with the value 'e.g. FB001'), 'Age' (empty), 'Weight (kg)' (empty), 'Wellness Goal' (a dropdown menu with 'General wellness' selected), and 'Intensity' (a dropdown menu with 'Low' selected). A blue 'Generate Plan' button is located below the form fields. At the bottom of the form, there is a small disclaimer: 'For safety, this demo accepts adults and only supports general wellness goals. It does not generate weight-loss prescriptions, calorie targets, or extreme workouts.'

Your Personalized Workout Plan

Your AI-generated fitness plan based on your information.

Name Gayathri	User ID FB001	Age 18	Weight 60.0 kg	Fitness Goal General wellness
Workout Intensity Low				

7-Day Workout Plan

7-DAY WELLNESS WORKOUT PLAN				
DAY 1 – FULL BODY				
Warm-up: 5–10 minutes of easy movement.				
Main workout:				
- Bodyweight squats – 2 sets				
- Wall push-ups – 2 sets				
- Glute bridges – 2 sets				
- Standing knee raises – 2 sets				
Cooldown: Gentle stretching and relaxed breathing.				
DAY 2 – UPPER BODY				
Warm-up: 5–10 minutes.				
Main workout:				
- Wall push-ups – 2 sets				
- Shoulder rolls – 2 sets				
- Arm raises – 2 sets				
- Standing punches – 2 sets				
Cooldown: Gentle upper-body stretching.				
DAY 3 – LOWER BODY				
Warm-up: 5–10 minutes.				
Main workout:				
- Chair squats – 2 sets				
- Glute bridges – 2 sets				
- Calf raises – 2 sets				
- Side leg raises – 2 sets				
Cooldown: Gentle lower-body stretching.				
DAY 4 – ACTIVE RECOVERY				
Warm-up: Easy walking for a few minutes.				
Main workout:				
- Gentle yoga stretches – 2 sets				
- Deep breathing – 2 sets				
- Neck rolls – 2 sets				
- Ankle rolls – 2 sets				
Cooldown: Relax and stretch your whole body.				
DAY 5 – FULL BODY				
Warm-up: 5–10 minutes.				
Main workout:				
- Jumping jacks – 2 sets				
- Lunges – 2 sets				
- Push-ups (knee or normal) – 2 sets				
- Plank – 2 sets				
Cooldown: Gentle stretching and relaxed breathing.				
DAY 6 – ACTIVE RECOVERY				
Warm-up: Easy walking for a few minutes.				
Main workout:				
- Gentle stretching – 2 sets				
- Breathing exercises – 2 sets				
- Shoulder and hip mobility – 2 sets				
- Light yoga poses – 2 sets				
Cooldown: Relax and stretch your whole body.				
DAY 7 – FULL BODY/RECOVERY				
Warm-up: 5–10 minutes.				
Main workout:				
- Squats – 2 sets				
- Push-ups – 2 sets				
- Glute bridges – 2 sets				
- Side leg raises – 2 sets				
Cooldown: Gentle stretching and deep breathing.				

Screenshot 4 - Feedback Page

Feedback

Share your feedback about your current workout plan. You can mention its effectiveness, difficulty, or your preferences.

User ID

FB001

Your Feedback

The workout plan is easy to understand and suitable for my daily routine. I would like slightly shorter activities and more rest between exercises.

Submit Feedback

Create Another Plan

Screenshot 5 - View All Users / Admin Page

All Users

[Home](#)

User ID	Name	Age	Weight	Goal	Intensity	Updated Plan?
FB006	hema	25	74.0 kg	General wellness	Low	Yes
FB004	Gayathri	20	60.0 kg	General wellness	Low	Yes
FB002	Banumathi	23	63.0 kg	General wellness	Medium	Yes
FB001	Gayathri	18	60.0 kg	General wellness	Low	Yes

FitBuddy is an educational software project, not medical advice.

13. Conclusion

FitBuddy demonstrates how generative AI can be combined with a web application to provide personalized fitness planning. The application collects user information and presents a structured workout plan based on the selected fitness goal and intensity. Its FastAPI backend, Jinja2 templates and database layer provide the basic structure needed for a complete web solution.

A key part of the project is personalization: the plan is intended to respond to the user's provided information rather than displaying one fixed plan. The project also provides a foundation for feedback-based refinement and administrative monitoring of stored user data.

Overall, the project gives the group practical experience in FastAPI, HTML/Jinja2 templates, database handling and integration with a generative AI service.